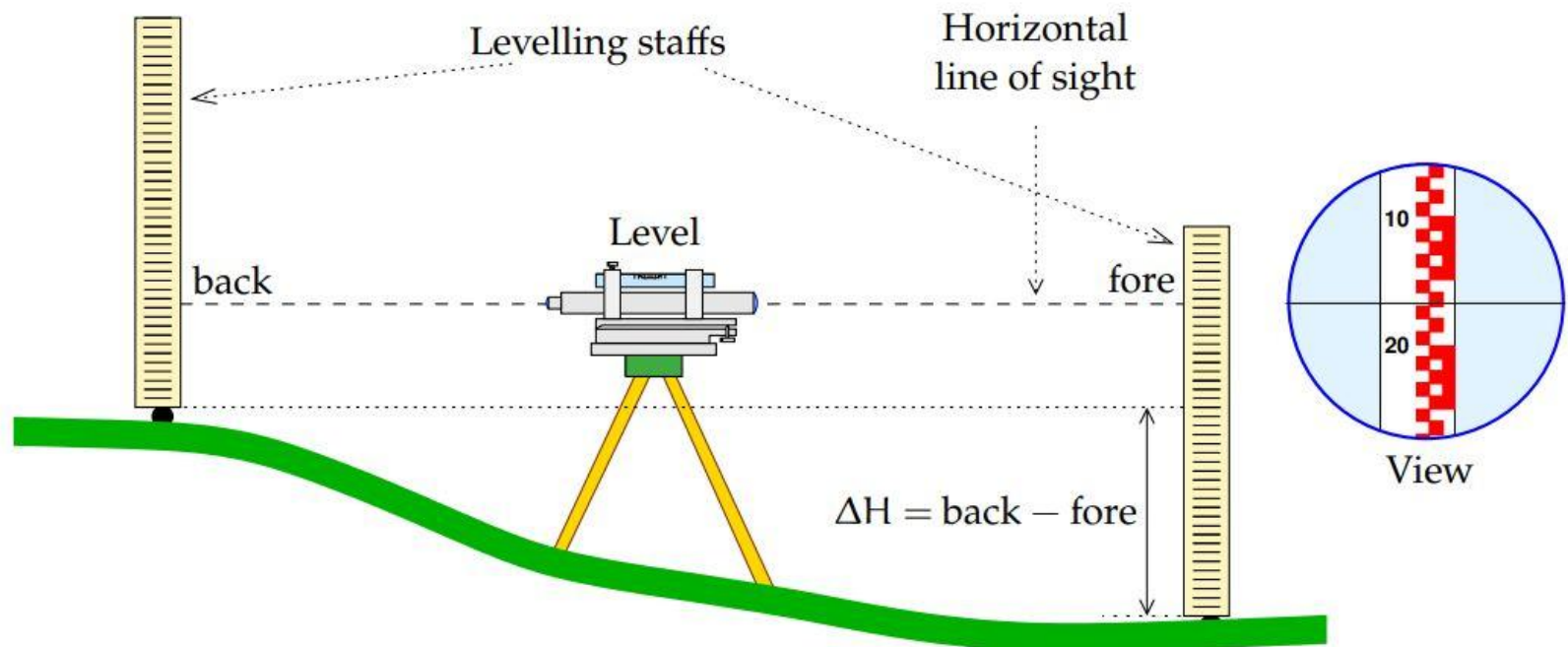


Levelling is a technique for determining height differences using a level (levelling instrument) and two rods or staffs.



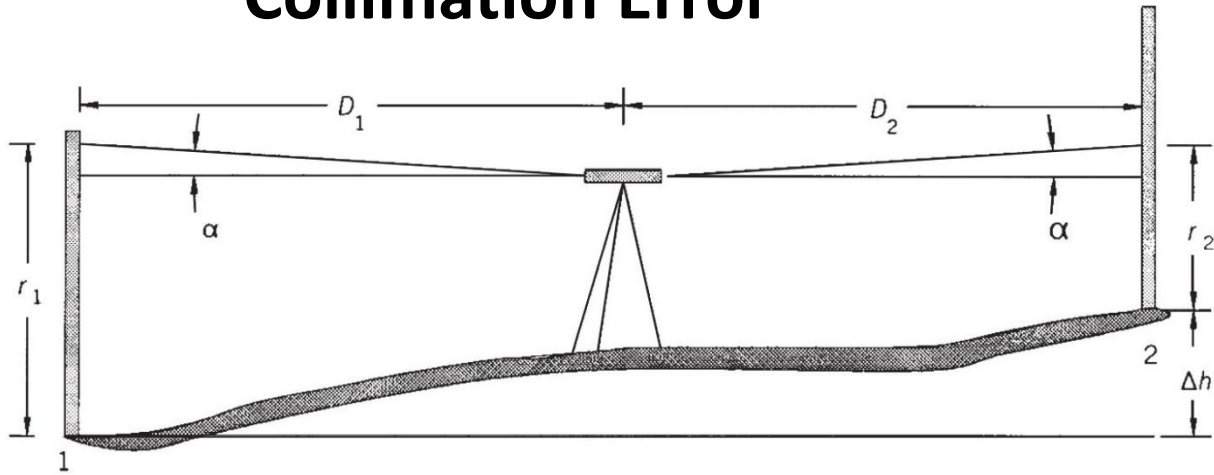
ERROR PROPAGATION IN ELEVATION DETERMINATION

The primary systematic errors include

- 1. EARTH CURVATURE,**
- 2. ATMOSPHERIC REFRACTION,**
- 3. INSTRUMENT MAL ADJUSTMENT.**

The effects of these systematic errors can be minimized by following proper field procedures, they can be modeled and corrected computationally.

Collimation Error



Collimation error in differential leveling.

$$e_c = D_1 \alpha - D_2 \alpha$$

$$e_c = \alpha [(D_1 - D_2) + (D_3 - D_4) + \dots + (D_{n-1} - D_n)]$$

$$e_c = \alpha (\sum D_{BS} - \sum D_{FS})$$

The collimation error determined from eqⁿ is treated as a corrⁿ

Subtracted from the observed elevation difference to obtain the corrected value.

Collimation Error Example

01	02	03
Given Data	Calculate Error	Corrected Result
Collimation error: 0.04 mm/m	$e_c = 0.00004(863 - 932) =$ -0.00276 m	$22.865 - (-0.00276)$ $= 22.868 \text{ m}$
Backsight sum: 863 m		
Foresight sum: 932 m		
Observed elevation: 22.865 m		

Curvature and Refraction

As the line of sight extends from an instrument, the level surface curves down and away. This condition always causes rod readings to be too high.

Similarly, as the line of sight extends from the instrument, refraction bends it toward the surface, causing readings to be too low.

The combined effect of Earth curvature and refraction on an individual sight always causes a rod reading to be too high by an amount approximated as

$$h_{CR} = CR \left(\frac{D}{1000} \right)^2$$

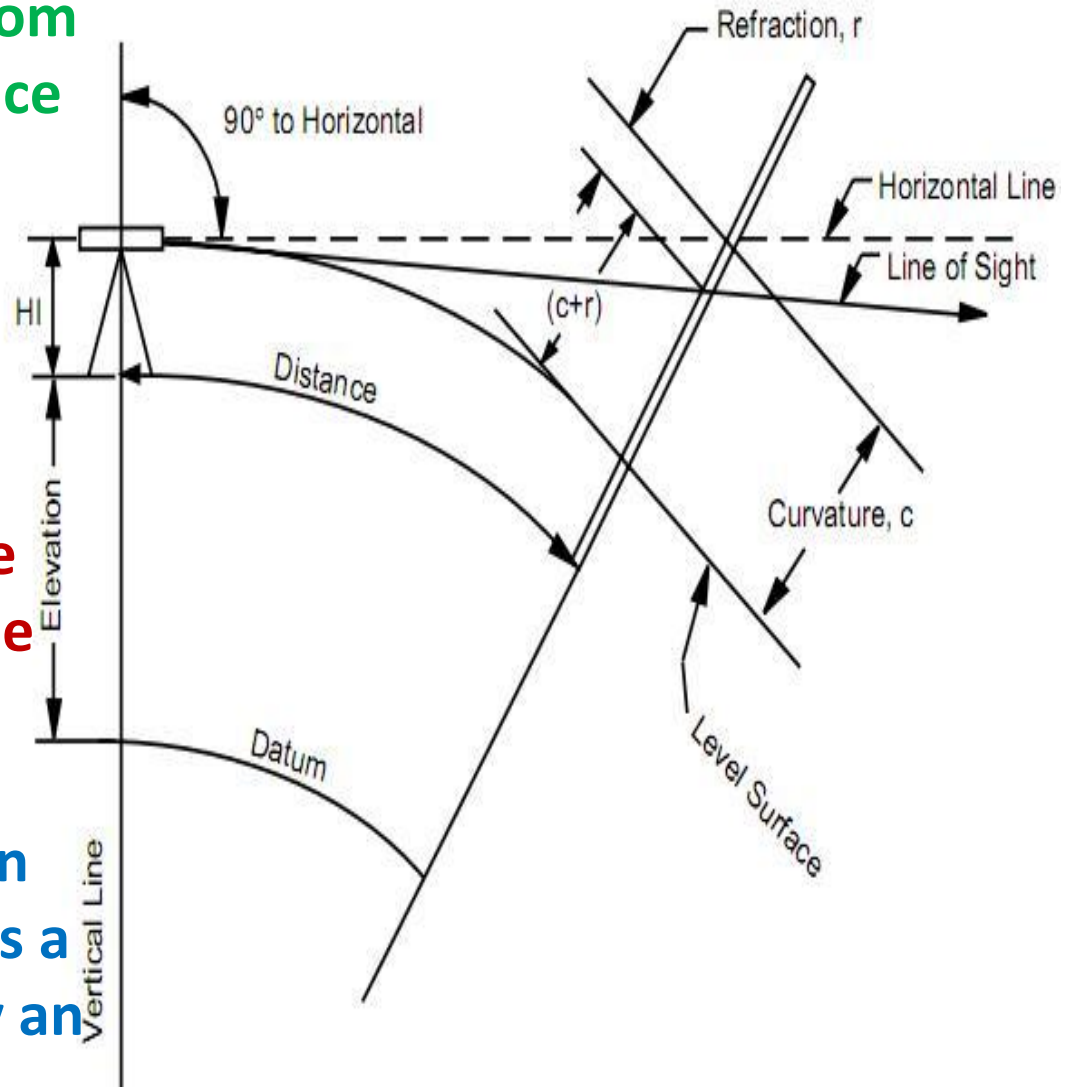
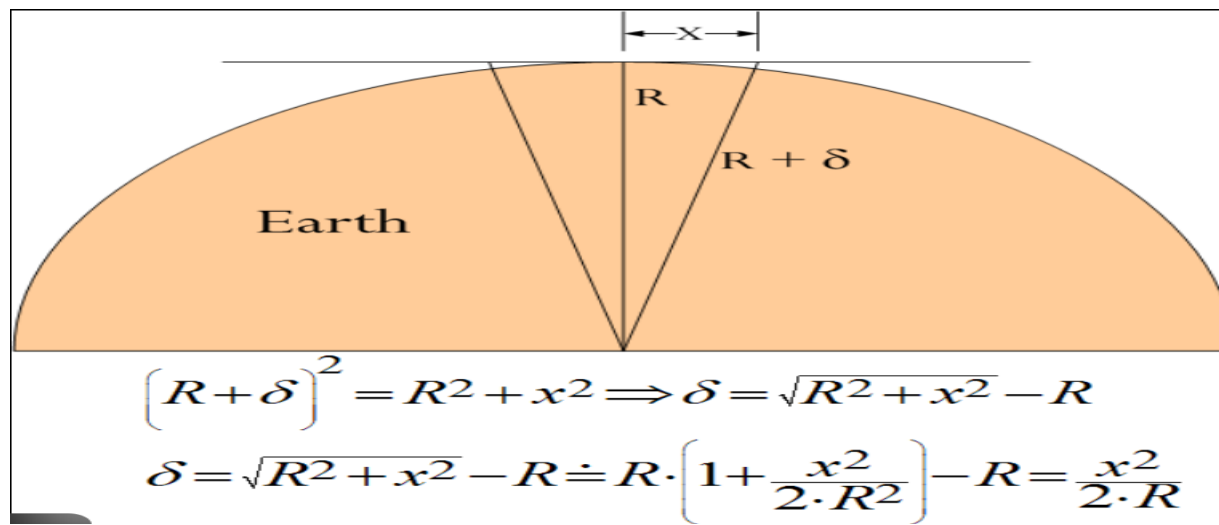


Figure 6-2. Curvature and refraction.



5. Substitute Earth's radius

Mean Earth radius:

$$R = 6\,370\,000 \text{ m}$$

$$C = \frac{d^2}{2 \times 6\,370\,000}$$

$$C = \frac{d^2}{12\,740\,000}$$

6. Convert distance into kilometres

Let:

$$d = 1000D \quad (\text{where } D \text{ is in km})$$

$$C = \frac{(1000D)^2}{12\,740\,000}$$

$$C = \frac{1\,000\,000D^2}{12\,740\,000}$$

$$C = 0.0785 D^2 \quad \text{meters}$$

Earth Curvature and Refraction

Curvature Effect

Level surface curves down and away from instrument

Always causes rod readings too high

Refraction Effect

Line of sight bends toward surface

Causes readings too low

Combined Formula

$$e_{CR} = \frac{CR}{1000^2} (D_1^2 - D_2^2)$$

$$CR = 0.0675 \text{ (meters)}$$

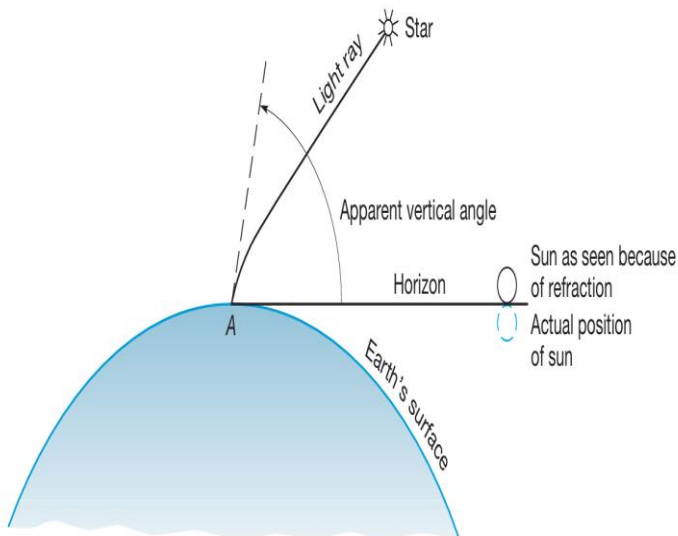
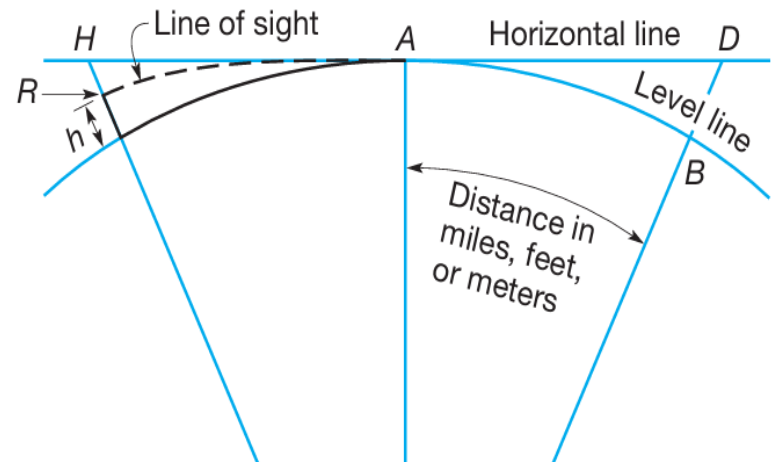


Figure 4.3
Refraction.



Curvature & Refraction Example

Problem Setup

Observed elevation difference:
1.256 m

Backsight distance: 100 m

Foresight distance: 20 m

Solution

$$e_{CR} = \frac{0.0675}{1000^2}(100^2 - 20^2) = 0.0006 \text{ m}$$

Corrected: 1.255 m

Key Principle

Keep backsight and foresight distances short and equal to minimize error

Sight lines must be ≥ 0.3 m above ground to avoid lower atmosphere refraction

9.2.3 Combined Effects of Systematic Errors on Elevation Differences

With reference to Figure 9.1, and by combining Equations (9.1) and (9.5), a corrected elevation difference, Δh , for one instrument setup is

$$\Delta h = (r_1 - r_2) - (D_1\alpha - D_2\alpha) - \frac{CR}{1000^2}(D_1^2 - D_2^2) \quad (9.9a)$$

where r_1 is the backsight rod reading, r_2 is the foresight rod reading, and the other terms are as defined previously. For a line of levels, the corrected elevation difference is

$$\Delta h = \sum r_{BS} - \sum r_{FS} - \alpha \left(\sum D_{BS} - \sum D_{FS} \right) - \frac{CR}{1000^2} \left(\sum D_{BS}^2 - \sum D_{FS}^2 \right)$$

RANDOM ERRORS IN DIFFERENTIAL LEVELING

The random errors in differential levelling occur in

INSTRUMENT LEVELLING

DISTANCE OBSERVATIONS

READING GRADUATED SCALES

These must be treated according to the theory of random errors.

The sizes of these errors are affected by atmospheric conditions, the quality of the optics of the telescope, the sensitivity of the level bubble or compensator, and the graduation scale on the rods.

Reading Errors

The estimated error in rod readings is usually expressed as a ratio of the estimated standard error in the rod reading per unit sight distance length. For example, if an observer's ability to read a rod is within $\sigma_{r/D} = \pm 0.00005$ m per m then $\sigma_{r/D}$ is $\pm 0.005/100 = \pm 0.00005$ ft/ft. Using this, rod reading errors for any individual sight distance D can be estimated as

$$\sigma_r = D\sigma_{r/D} \quad (9.10)$$

where $\sigma_{r/D}$ is the estimated error in the rod reading per unit length of sight distance and D the length of the sight distance.

Instrument Leveling Errors

The estimated error in leveling for an automatic compensator or level vial is generally given in the technical data for each instrument. For precise levels, this information is usually listed in arc seconds or as an estimated elevation error for a given distance. As an example, the estimated error may be listed as ± 1.5 mm/km, which corresponds to $\pm(1.5/1,000,000)\rho = \pm 0.3''$. A precise level has a compensator accuracy or setting accuracy between $\pm 0.1''$ and $\pm 0.2''$ usually, while for a less precise level the value may be as high as $\pm 10''$.

Rod Plumbing Error :

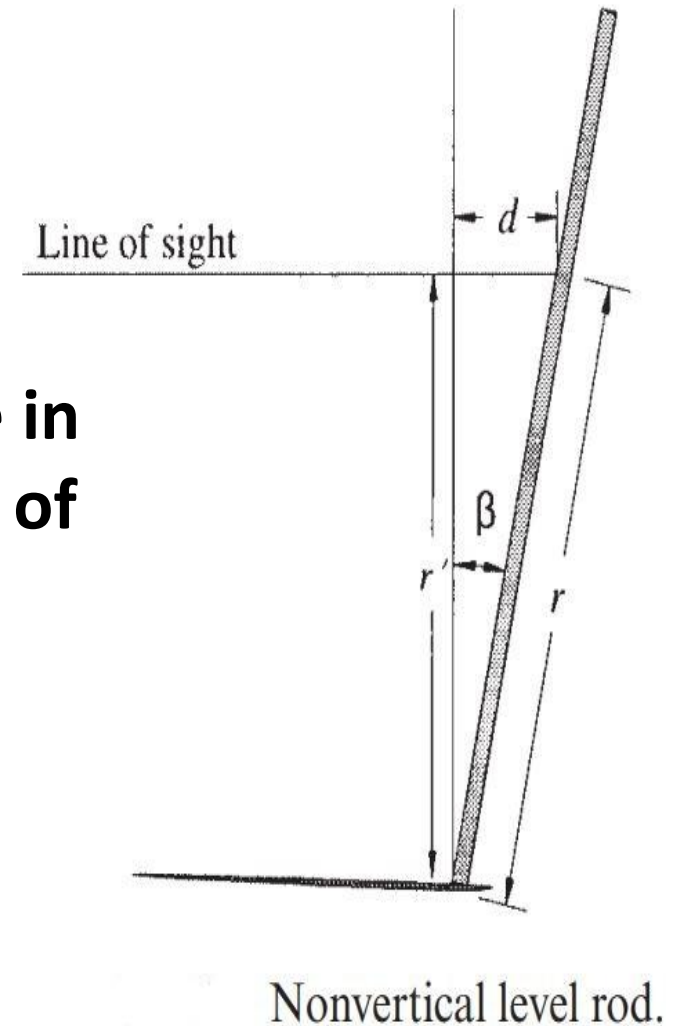
non vertical level rod always causes the reading to be too high,

this error will appear random in a leveling network due to its presence in all backsight and foresight distances of the network.

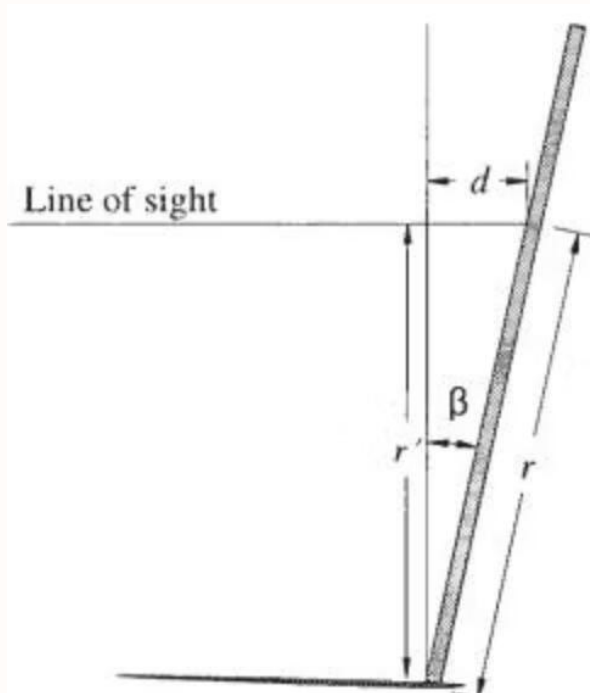
$$e_{LS} = r - r' = d^2 / 2r$$

$$d = r \sin \beta$$

$$e_{LS} = (r/2) \sin^2 \beta$$



Rod Plumbing Error Analysis



Error Characteristics

Rod out of plumb by angle β creates reading error

Backsight errors tend to cancel foresight errors

Example Calculation

Rod bubble: $\pm 5'$ of level

Rod reading: 4 m

$$e_{LS} = \frac{4}{2} \sin^2 5' = 0.004 \text{ mm}$$